

## DE KALB COUNTY AP, IN, USA

WMO: 720543

Lat: 41.300N

Lon: 85.067W

Elev: 268

StdP: 98.14

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00167

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -17.4      | -14.1 | -22.7                         | 0.5 | -15.2 | -19.9 | 0.7 | -12.8 | 10.2                  | -2.4 | 9.0 | -3.7 | 1.8                   | 250  | 0.467 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.2                   | 32.4            | 22.1 | 31.1 | 21.8 | 29.8 | 20.8 | 24.2                | 29.7 | 23.3 | 28.5 | 22.4 | 27.2 | 3.2                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 22.5                            | 17.7 | 27.3 | 22.1 | 17.3 | 26.7 | 21.1 | 16.3 | 25.6 | 74.3          | 30.0 | 70.9 | 28.4 | 67.1 | 27.0 | 27.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 8.6           | 7.7  | 6.9 | DB                         | -21.1 | 34.7               | 4.0 | 2.8                                                | -23.9 | 36.7       | -26.3 | 38.3       | -28.5 | 39.9       | -31.4 | 41.9 | (4) |
| (5)               |      |     | WB                         | -21.5 | 25.3               | 4.0 | 1.1                                                | -24.4 | 26.1       | -26.7 | 26.8       | -29.0 | 27.4       | -31.9 | 28.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.3        | -3.8  | -2.2  | 3.4   | 9.4   | 16.4  | 21.0  | 22.7  | 21.5  | 18.2  | 11.4  | 4.6   | -0.3  | (6)  |      |
|                                                                        | DBStd       | 10.49       | 6.24  | 6.08  | 5.72  | 4.84  | 4.83  | 3.36  | 3.31  | 2.87  | 4.09  | 4.90  | 5.07  | 5.20  | (7)  |      |
|                                                                        | HDD10.0     | 1603        | 429   | 343   | 219   | 68    | 6     | 0     | 0     | 0     | 1     | 43    | 174   | 320   | (8)  |      |
|                                                                        | HDD18.3     | 3377        | 687   | 574   | 465   | 271   | 95    | 13    | 3     | 7     | 52    | 222   | 412   | 576   | (9)  |      |
|                                                                        | CDD10.0     | 1697        | 1     | 2     | 13    | 49    | 205   | 329   | 395   | 356   | 247   | 85    | 12    | 2     | (10) |      |
|                                                                        | CDD18.3     | 429         | 0     | 0     | 1     | 2     | 36    | 93    | 140   | 104   | 48    | 6     | 0     | 0     | (11) |      |
|                                                                        | CDH23.3     | 3877        | 0     | 0     | 7     | 31    | 367   | 784   | 1329  | 880   | 417   | 60    | 1     | 0     | (12) |      |
| (13)                                                                   | CDH26.7     | 1212        | 0     | 0     | 0     | 1     | 103   | 241   | 487   | 242   | 131   | 8     | 0     | 0     | (13) |      |
| (14)                                                                   | Wind        | WSAvg       | 2.6   | 3.3   | 3.3   | 3.2   | 3.5   | 2.6   | 2.2   | 1.8   | 1.6   | 1.9   | 2.5   | 2.8   | 2.9  | (14) |
| (15) Precipitation                                                     | PrecAvg     | 965         | 63    | 48    | 62    | 89    | 106   | 93    | 106   | 106   | 78    | 76    | 75    | 64    | (15) |      |
|                                                                        | PrecMax     | 1278        | 121   | 142   | 119   | 154   | 188   | 152   | 183   | 274   | 174   | 221   | 166   | 150   | (16) |      |
|                                                                        | PrecMin     | 757         | 22    | 11    | 17    | 8     | 17    | 37    | 41    | 42    | 10    | 6     | 19    | 21    | (17) |      |
|                                                                        | PrecStd     | 123         | 31    | 31    | 22    | 37    | 51    | 36    | 39    | 53    | 48    | 52    | 30    | 32    | (18) |      |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 13.7  | 17.4  | 25.1  | 26.4  | 32.0  | 33.0  | 36.1  | 32.5  | 32.7  | 27.7  | 21.3  | 16.2  | (19) |      |
|                                                                        |             | MCWB        | 11.9  | 10.6  | 17.0  | 15.2  | 20.5  | 21.7  | 23.4  | 21.8  | 21.8  | 18.9  | 13.6  | 13.6  | (20) |      |
|                                                                        | 2%          | DB          | 10.2  | 12.8  | 18.6  | 23.7  | 29.8  | 31.3  | 32.5  | 31.1  | 30.9  | 25.1  | 17.7  | 12.7  | (21) |      |
|                                                                        |             | MCWB        | 9.0   | 10.2  | 13.2  | 14.5  | 18.9  | 21.4  | 22.7  | 21.7  | 21.9  | 17.7  | 12.4  | 10.8  | (22) |      |
|                                                                        | 5%          | DB          | 7.3   | 9.1   | 15.3  | 21.2  | 27.5  | 29.8  | 31.2  | 29.1  | 27.6  | 22.4  | 15.2  | 9.9   | (23) |      |
|                                                                        |             | MCWB        | 5.7   | 6.9   | 10.5  | 13.2  | 18.4  | 20.8  | 22.3  | 20.7  | 20.3  | 16.2  | 10.5  | 8.1   | (24) |      |
|                                                                        | 10%         | DB          | 4.0   | 5.9   | 12.6  | 17.8  | 25.2  | 27.6  | 29.8  | 27.6  | 25.9  | 20.0  | 12.5  | 7.0   | (25) |      |
|                                                                        |             | MCWB        | 2.3   | 3.3   | 8.8   | 11.5  | 17.3  | 19.4  | 21.3  | 20.1  | 18.5  | 15.1  | 8.8   | 5.2   | (26) |      |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 12.0  | 12.6  | 17.1  | 17.5  | 22.6  | 24.8  | 25.8  | 24.4  | 24.4  | 21.8  | 15.9  | 14.2  | (27) |      |
|                                                                        |             | MCDB        | 13.8  | 14.8  | 24.3  | 22.9  | 28.5  | 30.9  | 31.6  | 29.6  | 30.0  | 25.8  | 19.8  | 16.2  | (28) |      |
|                                                                        | 2%          | WB          | 8.9   | 10.1  | 14.4  | 15.7  | 20.9  | 23.1  | 24.7  | 23.3  | 22.8  | 19.1  | 13.3  | 11.4  | (29) |      |
|                                                                        |             | MCDB        | 10.0  | 13.0  | 17.9  | 20.5  | 26.4  | 28.9  | 30.3  | 28.1  | 27.9  | 22.6  | 16.9  | 12.4  | (30) |      |
|                                                                        | 5%          | WB          | 5.7   | 7.1   | 12.0  | 14.3  | 19.6  | 22.1  | 23.7  | 22.4  | 21.8  | 17.3  | 11.3  | 7.8   | (31) |      |
|                                                                        |             | MCDB        | 7.3   | 9.1   | 14.1  | 19.5  | 25.3  | 27.3  | 28.9  | 26.7  | 26.2  | 20.2  | 14.7  | 9.0   | (32) |      |
|                                                                        | 10%         | WB          | 2.8   | 3.7   | 9.3   | 12.8  | 18.7  | 21.0  | 22.7  | 21.6  | 20.5  | 15.7  | 8.9   | 5.2   | (33) |      |
|                                                                        |             | MCDB        | 4.1   | 5.7   | 13.3  | 18.1  | 23.6  | 25.1  | 27.5  | 25.7  | 23.9  | 19.7  | 12.6  | 6.8   | (34) |      |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.5   | 7.9   | 9.3   | 11.3  | 11.6  | 11.0  | 11.2  | 11.4  | 11.4  | 10.5  | 8.4   | 6.5   | (35) |      |
|                                                                        |             | MCDBR       | 9.7   | 11.9  | 12.9  | 15.2  | 14.2  | 13.1  | 12.5  | 13.0  | 13.2  | 13.8  | 12.8  | 10.2  | (36) |      |
|                                                                        | 5% WB       | MCWBR       | 8.5   | 9.2   | 8.9   | 8.6   | 6.8   | 5.5   | 4.7   | 5.3   | 5.9   | 7.4   | 8.4   | 8.6   | (37) |      |
|                                                                        |             | MCWBR       | 9.4   | 11.8  | 11.3  | 12.8  | 11.7  | 10.8  | 10.6  | 9.8   | 11.0  | 11.4  | 11.6  | 9.7   | (38) |      |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.300 | 0.333 | 0.350 | 0.400 | 0.437 | 0.441 | 0.437 | 0.441 | 0.400 | 0.352 | 0.320 | 0.302 | (40) |      |
|                                                                        |             | taud        | 2.393 | 2.290 | 2.343 | 2.217 | 2.160 | 2.191 | 2.218 | 2.193 | 2.287 | 2.416 | 2.456 | 2.426 | (41) |      |
|                                                                        | Edn at Noon | Edn at Noon | 852   | 868   | 897   | 870   | 843   | 838   | 837   | 821   | 832   | 840   | 822   | 818   | (42) |      |
|                                                                        |             | Edh at Noon | 81    | 105   | 112   | 137   | 148   | 144   | 139   | 138   | 118   | 92    | 76    | 73    | (43) |      |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.54  | 2.43  | 3.51  | 4.63  | 5.35  | 6.01  | 6.07  | 5.33  | 4.36  | 2.89  | 1.84  | 1.30  | (44) |      |
|                                                                        | RadStd      | RadStd      | 0.16  | 0.33  | 0.32  | 0.41  | 0.49  | 0.40  | 0.33  | 0.27  | 0.36  | 0.27  | 0.26  | 0.15  | (45) |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (56 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page